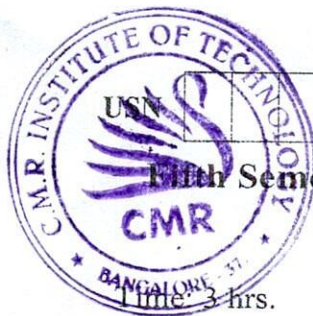




CBCS SCHEME

BCS503

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Theory of Computation

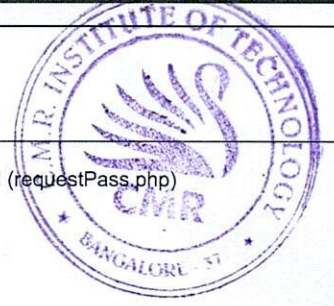
Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C															
Q.1	a.	Define the following with example i) Language ii) String iii) Power of an alphabet.	3	L1	CO1															
	b.	Define DFA. Draw a DFA to accepts. i) The set of all strings that contain a substring aba. ii) To accept the strings of a's and b's that contain not more than three b's. iii) $L = \{w \in \{a, b\}^* : \text{No 2 consecutive characters are same in } w\}$.	10	L3	CO1															
	c.	Convert the following NFA to DFA. <table><tr><td></td><td>0</td><td>1</td></tr><tr><td>$\rightarrow$ p</td><td>{p, q}</td><td>{p}</td></tr><tr><td>q</td><td>{r}</td><td>{r}</td></tr><tr><td>r</td><td>{s}</td><td>ϕ</td></tr><tr><td>* s</td><td>{s}</td><td>{s}</td></tr></table>		0	1	$\rightarrow$ p	{p, q}	{p}	q	{r}	{r}	r	{s}	ϕ	* s	{s}	{s}	7	L2	CO1
	0	1																		
$\rightarrow$ p	{p, q}	{p}																		
q	{r}	{r}																		
r	{s}	ϕ																		
* s	{s}	{s}																		
OR																				
Q.2	a.	Define the following with example : i) Alphabet ii) Reversal of string iii) Concatenation of Languages.	3	L1	CO1															
	b.	Design a DFA for the Language : $L = \{w \in \{0, 1\}^* : w \text{ is a string divisible by } 5\}$.	7	L3	CO1															
	c.	Define NFA. Obtain an ϵ -NFA which accepts strings consisting of 0 or more a's , followed by 0 or more b's followed by 0 or more C's. Also convert it to DFA.	10	L2	CO1															
Module – 2																				
Q.3	a.	Define Regular expression. Write the regular expression for the following languages : i) Strings of a's and b's starting with a and ending with b. ii) Set of strings that consists of alternating 0's and 1's. iii) $L = \{a^n b^m, (n + m) \text{ is even}\}$. iv) $L = \{w : w \text{ mod } 3 = 0, \text{ where } w \in \{a, b\}^*\}$.	10	L2	CO2															

b. Minimize the following finite automata using Table filling algorithm :		10	L2	CO2
$ \begin{array}{c cc} \delta & a & b \\ \hline \rightarrow A & B & A \\ B & A & C \\ C & D & B \\ * D & D & A \\ E & D & F \\ F & G & E \\ G & F & G \\ H & G & D \end{array} $				
OR				
Q.4	a. Construct ε - NFA for the following Regular expression : i) $(0+1)01(1+0)$ ii) $1(0+1)^*0$ iii) $(0+1)^*011^*$	6	L1	CO2
	b. Obtain the Regular expression that denotes the language accepted by Fig. Q4(b). Fig. Q4(b) Using Kleene's theorem.	6	L3	CO2
	c. State the Pumping Lemma for the Regular Languages. And also prove that the following languages are not regular. i) $L = \{0^n 1^m \mid n \leq m\}$ ii) $L = \{0^n 1^m 2^n \mid n, m \geq 1\}$.	8	L1	CO2
Module - 3				
Q.5	a. Design CFG for the following languages : i) $L = \{a^n b^{n+3}, n \geq 0\}$ ii) $L = \{a^i b^j c^k, j = i + k, i \geq 0, k \geq 0\}$ iii) $L = \{w \mid w \bmod 3 > 0 \text{ where } w \in \{a\}^*\}$ iv) $L = \{a^m b^n \mid m \neq n\}$ v) Palindromes over 0 and 1.	10	L3	CO3
	b. Consider the grammar G with productions. $S \rightarrow A b B / A / B$; $A \rightarrow aA / \varepsilon$; $B \rightarrow aB / bB / \varepsilon$. Obtain LMD, RMD and parse tree for the string aabab. Is the given grammar ambiguous?	10	L2	CO3
OR				
Q.6	a. Define the following with example : i) Context free grammar ii) Left most Derivation iii) Parse tree iv) Ambiguous grammar.	4	L1	CO3
	b. Design PDA for the language : $L = \{a^i b^j c^k \mid i + k = j, i \geq 0, k \geq 0\}$ and show the moves made by the PDA for the string aabbbc.	10	L3	CO3

	c.	Convert the following CFG's to PDA : $S \rightarrow aA$; $A \rightarrow aABC / bB / a$; $B \rightarrow b$; $C \rightarrow c$.	6	L2	CO3
Module – 4					
Q.7	a.	Define CNF. Convert the following CFG to CNF. $E \rightarrow E + T / T$ $T \rightarrow T * F / F$ $F \rightarrow (E) / I$ $I \rightarrow Ia / Ib / a / b$.	10	L2	CO4
	b.	Show that $L = \{0^n 1^n 2^n / n \geq 1\}$ is not context free.	4	L2	CO4
	c.	Prove that the family of context free languages is closed under union and concatenation.	6	L1	CO4
OR					
Q.8	a.	Define Greibach Normal Form. Convert the following CFG to GNF. $S \rightarrow AB$; $A \rightarrow aA / bB / b$; $B \rightarrow b$.	6	L2	CO4
	b.	Consider the following CFG : $S \rightarrow ABC / BaB$ $A \rightarrow aA / BaC / aaa$ $B \rightarrow bBb / a / D$ $C \rightarrow CA / AC$ $D \rightarrow \epsilon$ i) What are useless symbols? ii) Eliminate ϵ - productions, Unit productions and useless symbols from the grammar.	10	L3	CO4
	c.	Prove that the following languages are not context free. i) $L = \{a^i / i \text{ is prime}\}$ ii) $L = \{a^{n^2} / n \geq 1\}$.	4	L2	CO3
Module – 5					
Q.9	a.	Define a Turing machine and explain with neat diagram, the working of a basic Turing machine.	6	L1	CO4
	b.	Design a Turing machine to accept the language, $L = \{a^n b^n c^n / n \geq 1\}$. Draw the transition diagram and show the moves for the string aabbcc.	14	L4	CO4
OR					
Q.10	a.	Design a Turing machine to accept palindrome over $\{a, b\}$ and draw the transition diagram.	12	L4	CO5
	b.	Write a short notes on : i) Recursively Enumerable Language. ii) Multitape Turing Machine.	8	L1	CO5



Current Session : 02:00:00

02:00:00 Session is in progress.

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Feedback/Remarks

Date:17-01-2025 | Session:02:00:00

For Subject Code : BCS503

1. Q. 1b. There to be read as **three**
2. Q. 3a. bm to b^m
3. Q. 4c. note to be read as **not**
4. Q. 5c. palinderomes to be read as **palindromes**
5. Q. 7b. $2n$ to 2^n
6. Q. 8c. ai to a^i

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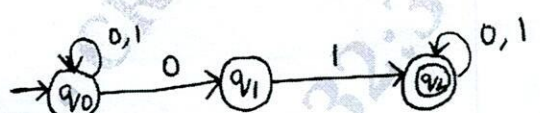


Fifth Semester B.E./B.Tech. Degree Examination, June/July 2025
Theory of Computation

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.

Module - 1				M	L	C
Q.1	a.	Define the following with an example: i) Alphabet ii) Power of an alphabet iii) String iv) String concatenation v) Language		5	L1	CO1
	b.	Define Deterministic Finite Automata (DFA) and the language accepted by it.		5	L1	CO1
	c.	Design DFA to accept the following languages: i) $L = \{W \in \{0, 1\}^* : W \text{ has } 001 \text{ as a substring}\}$ ii) $L = \{W \in \{0, 1\}^* : W \bmod 3 = 0\}$		10	L3	CO1
OR						
Q.2	a.	Convert the following NFA to DFA  Fig.Q.2(a)		8	L2	CO1
	b.	Convert the following ϵ -NFA to DFA  Fig.Q.2(b) and define ϵ -NFA		12	L2	CO1
Module - 2						
Q.3	a.	Define regular expression. Write the regular expression for the following languages: i) Representing for strings of a and b's having odd length. ii) To accept 10 as substring over an alphabet $\Sigma = \{0, 1\}$		10	L2	CO2
	b.	State and prove pumping Lemma for regular languages.		10	L2	CO2

OR						
Q.4	a.	Prove that regular languages are closed under complementation and intersection.		10	L2	CO2
	b.	i) Obtain NFA (Non deterministic finite automata) for the regular expression $(a + b)^* abb$. ii) Obtain NFA for the regular expression $(a^* + ab)(a + b)^*$		6	L2	CO2
	c.	Write the applications of regular expression.		4	L2	CO2
Module - 3						
Q.5	a.	Define context free grammar. Write the CFG for the following languages: i) $L = \{a^n b^n c^m : n \geq 0, m \geq 0\}$ ii) $L = \{w \in \{a, b\}^* : n_a(w) = n_b(w)\}$		10	L2	CO2
	b.	i) Define ambiguous grammar with suitable example. ii) Consider the grammar $E \rightarrow + EE / * EE / - EE / x / y$ Find the left most derivation, right most derivation and parse tree for the string $++* -xyxy$.		10	L2	CO2
OR						
Q.6	a.	Define PDA (Push Down Automata). Design a PDA to accept the following language: $L = \{a^n b^n : n \geq 0\}$. Draw the transition diagram and show that instantaneous description for the string aaabbb.		10	L3	CO3
	b.	Convert the following CFG to PDA: i) $E \rightarrow E + E \mid E * E \mid id$ ii) $E \rightarrow I \mid E * E \mid (E)$ $I \rightarrow id$		6	L2	CO3
	c.	Discuss the language accepted by PDA.		4	L1	CO3
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Module - 4						
Q.7	a.	Convert the following grammar to CNF (Chomsky Normal Form) $S \rightarrow ASB / \epsilon$ $A \rightarrow aAS / a$ $B \rightarrow SbS / A / bb$ and define CNF		10	L2	CO3
	b.	State and prove pumping Lemma for context free languages.		10	L2	CO3

OR

Q.8	a.	What are useless and ϵ productions? Eliminate ϵ , unit and useless productions from the following grammar: $A \rightarrow bA / Bba / aa$ $B \rightarrow aBa / b / D$ $C \rightarrow CA / AC / B$ $D \rightarrow a / \epsilon$	10	L3	CO3
	b.	Prove that the family of context free languages is closed under union concatenation and star closure.	10	L2	CO3

Module – 5

Q.9	a.	Define a Turing Machine. Explain the working and variants of Turing machine.	10	L1	CO4
	b.	Design a Turing machine to accept $L = \{a^n b^n c^n \mid n \geq 0\}$. Draw the transition diagram. Show the moves made for string aabbcc.	10	L3	CO4

OR

Q.10	a.	Explain language acceptability and design of Turing Machines (Steps).	10	L2	CO5
	b.	Explain the following: i) Programming techniques for turing machines ii) Undecidability problem.	10	L2	CO5

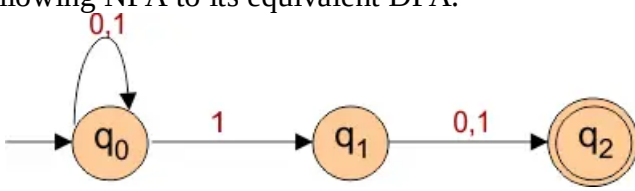
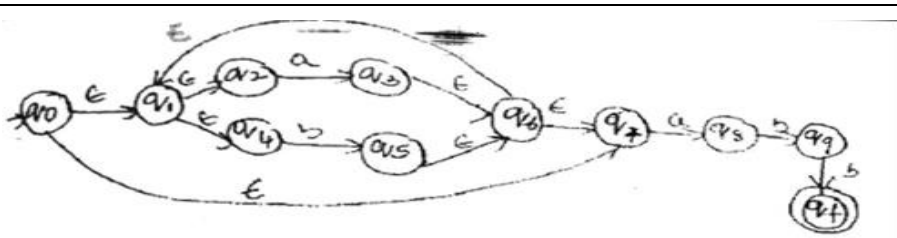
Model Question Paper with effect from 2024-25 (CBCS Scheme)

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Fifth Semester B.E. Degree Examination
Theory of Computation

TIME: 03 Hours**Max. Marks: 100**Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

Module 1				BL	Marks																				
Q.01	a	i. Design DFA to accept strings of a's and b's where language $L=\{ W \bmod 5= W \bmod 4\}$ ii. Design DFA to Accept stings of 0's and 1's $L=\{ \text{ starting with 01 or starting with 10}\}$	L3	10																					
	b	Define NFA. Convert the following NFA to its equivalent DFA. 	L3	10																					
OR																									
Q.02	a	 Find E-Closure of all the states Convert following NFA to DFA	L3	10																					
	b	1. Define Alphabets Strings and Languages. ii. Construct a DFA to accept strings of 0's and 1's starting with at least two 0's and ending with at least two 1's.	L2	10																					
Module-2																									
Q. 03	a	i. Define Regular Expressions. ii. Obtain RE to accept strings of a's and b's whose second symbol from the right end is a iii. Obtain RE to accept words with two or more letters but beginning and ending with the same letter where {a, b} are inputs. iv. Obtain NFA which accepts strings of a's and b's starting with the string ab v. Obtain NFA for the Regular Expression $(a+b)^*aa(a+b)^*$	L3	10																					
	b	Minimize the following DFA using Table filling Algorithm. <table><tr><td></td><td>0</td><td>1</td></tr><tr><td>→ A</td><td>B</td><td>A</td></tr><tr><td>B</td><td>A</td><td>C</td></tr><tr><td>C</td><td>D</td><td>B</td></tr><tr><td>* D</td><td>D</td><td>A</td></tr><tr><td>E</td><td>D</td><td>F</td></tr><tr><td>F</td><td>G</td><td>E</td></tr></table>		0	1	→ A	B	A	B	A	C	C	D	B	* D	D	A	E	D	F	F	G	E	L3	10
	0	1																							
→ A	B	A																							
B	A	C																							
C	D	B																							
* D	D	A																							
E	D	F																							
F	G	E																							

			G	F	G			
			H	G	D			
OR								
Q.04	a	i.	State and prove pumping lemma Theorem				L2	10
		ii.	Show that , $L= \{WW^R \mid W \in \{a,b\}^*\}$ is not regular					
	b	Show that regular languages are closed under					L3	10
		i.	Union,concatenation and Kleens star					
		ii.	Intersection and Difference					
Module-3								
Q. 05	a	i.	Obtain the unambiguous grammar for the grammar shown and get the derivation for the expression (a+b)*(a-b) . $E \rightarrow E+E \mid E-E$ $E \rightarrow E * E \mid E / E$ $E \rightarrow (E) \mid I$ $I \rightarrow a b c$				L3	10
		ii.	Consider the following grammar $S \rightarrow AbB$ $A \rightarrow aA \mid \epsilon$ $B \rightarrow aB \mid bB \mid \epsilon$ Give LMD and RMD and Parse tree for the string “aaabab “					
	b	a) Design a PDA for accepting a palindrome $L(M) = \{WCW^R \mid W \in (a+b)^*\}$ where W^R is reverse of W and show ID for the string “aab”					L3	10
OR								
Q. 06	a	Construct CFG for the following languages i) $L=\{0^{2n}1^m \mid n \geq 0, m \geq 0\}$ ii) $L= \{WW^R \mid W \in \{a,b\}^*\}$ iii) $L= \{ w \mid w \in \{0,1\}^* \text{ with atleast one occurance of “101” } \}$ iv) $L= \{ \text{ strings of a’s and b’s with equal number of a’s and b’s } \}$ V) Obtain a grammar to generate a language of strings 0’s and 1’s haing a substring 000					L2	10
	b	Define with example i. Grammar ii. Derivation iii. Leftmost and Rightmost derivation iv. Ambiguous grammar v. Parse tree					L3	10
Module-4								
Q. 07	a	$S \rightarrow a \mid aA \mid B$ $A \rightarrow aBB \mid \epsilon$ $B \rightarrow Aa \mid b$ Convert the above grammar to CNF					L3	10
	b	$S \rightarrow Sa \mid Sb \mid aA \mid B$ $A \rightarrow Ab \mid aBB \mid a \mid b$ $B \rightarrow Ba \mid bb$ Eliminate left recursion.					L3	10

OR					
Q. 08	a	$A \rightarrow BC$ $B \rightarrow CA a$ $B \rightarrow AB b$ Convert the above grammar to GNF	L3		10
	b	Show that CFL is not closed under union , Concatenation , star and complimentation	L3		10
Module-5					
Q. 09	a	i. Explain The Church-Turing machine with neat a diagram ii. Explain Multiple TM with a neat diagram	L2		10
	b	Define Turing Machine (TM). Design a TM for language $L=\{0^n 1^n \mid n \geq 0\}$ Show that the string 0011 is accepted	L3		10
OR					
Q. 10	a	Define Turing Machine (TM). Design a TM for language $L=\{1^n 2^n 3^n \mid n \geq 1\}$ Show that the string 111222333 is accepted	L4		12
	b	Demonstrate the model of Linear bound automata(LBA) with neat diagram	L3		08

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026
Theory of Computation

Time: 3 hrs.

Max. Marks: 100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.**

Module – 1			M	L	C																											
Q.1	a.	Define the following with example. i) Alphabet ii) String iii) Power of an Alphabet iv) Language v) Problem	05	L2	CO1																											
	b.	Design DFA for the following languages. i) DFA to accept string of 0's, 1's and 2's beginning with a '0' followed by odd number of 1's and ending with a '2'. ii) $L = \{ W \in \{0,1\}^* : w \text{ does not have } 001 \text{ as a substring} \}$	08	L3	CO1																											
	c.	Convert the following NFA to DFA. 	07	L3	CO1																											
OR																																
Q.2	a.	Design an NFA to recognize the following set of strings 0101, 101 and 011.	05	L2	CO1																											
	b.	Design an DFA to accept binary numbers divisible by 5.	08	L3	CO1																											
	c.	Obtain DFA for the following ϵ – NFA <table><tr><th>δ</th><th>ϵ</th><th>a</th><th>b</th><th>c</th></tr><tr><td>$\rightarrow p$</td><td>ϕ</td><td>$\{p\}$</td><td>$\{q\}$</td><td>$\{r\}$</td></tr><tr><td>q</td><td>$\{p\}$</td><td>$\{q\}$</td><td>$\{r\}$</td><td>ϕ</td></tr><tr><td>$*r$</td><td>$\{q\}$</td><td>$\{r\}$</td><td>ϕ</td><td>$\{p\}$</td></tr></table>	δ	ϵ	a	b	c	$\rightarrow p$	ϕ	$\{p\}$	$\{q\}$	$\{r\}$	q	$\{p\}$	$\{q\}$	$\{r\}$	ϕ	$*r$	$\{q\}$	$\{r\}$	ϕ	$\{p\}$	07	L3	CO1							
δ	ϵ	a	b	c																												
$\rightarrow p$	ϕ	$\{p\}$	$\{q\}$	$\{r\}$																												
q	$\{p\}$	$\{q\}$	$\{r\}$	ϕ																												
$*r$	$\{q\}$	$\{r\}$	ϕ	$\{p\}$																												
Module – 2																																
Q.3	a.	Write the Regular Expression for the following languages. i) $L = \{a^n b^m \mid n \geq 4, m \leq 3\}$ ii) RE to accept words with two or more letters but beginning and ending with the same letter where $\Sigma = \{a, b\}$	06	L3	CO2																											
	b.	Obtain ϵ – NFA for given Regular Expression $(a + b)^* aa (a + b)^*$	06	L3	CO2																											
	c.	Find the minimized DFA of the following <table><tr><th>δ</th><th>a</th><th>b</th></tr><tr><td>$\rightarrow A$</td><td>B</td><td>F</td></tr><tr><td>B</td><td>G</td><td>C</td></tr><tr><td>$*C$</td><td>A</td><td>C</td></tr><tr><td>D</td><td>C</td><td>G</td></tr><tr><td>E</td><td>H</td><td>F</td></tr><tr><td>F</td><td>C</td><td>G</td></tr><tr><td>G</td><td>G</td><td>E</td></tr><tr><td>H</td><td>G</td><td>C</td></tr></table>	δ	a	b	$\rightarrow A$	B	F	B	G	C	$*C$	A	C	D	C	G	E	H	F	F	C	G	G	G	E	H	G	C	08	L3	CO2
δ	a	b																														
$\rightarrow A$	B	F																														
B	G	C																														
$*C$	A	C																														
D	C	G																														
E	H	F																														
F	C	G																														
G	G	E																														
H	G	C																														
1 of 2																																

OR

Q.4	a.	State and prove pumping theorem for regular languages. Show that the language $L = \{a^n b^n \mid n \geq 0\}$ is not regular.	06	L3	CO2
	b.	Convert the following FA to RE using state elimination method. 	06	L3	CO2
	c.	Prove that the Regular Languages are closed under : i) Union ii) Complementation iii) Intersection iv) Difference	08	L3	CO2

Module – 3

Q.5	a.	Design CFG for the following Languages. i) $L = \{ ww^R \mid w \in \{a, b\}^* \}$ ii) $L = \{ 0^m 1^m 2^n \mid m \geq 1 \text{ and } n \geq 0 \}$ iii) $L = \{ a^n b^{n+3} \mid n \geq 0 \}$	10	L3	CO3
	b.	Consider the grammar G with productions $S \rightarrow aSbS \mid bSaS \mid \epsilon$ Obtain LMD, RMD and parse tree for the string aababb. Is the grammar ambiguous.	10	L3	CO3

OR

Q.6	a.	Obtain PDA to accept the language $L = \{ wCw^R \mid W \in (a + b)^* \}$ and show the moves made by the PDA for the string aabCbaa.	10	L3	CO3
	b.	Convert the following CFG to PDA $S \rightarrow aABC$ $A \rightarrow aB a$ $B \rightarrow bA b$ $C \rightarrow a$	10	L3	CO3

Module – 4

Q.7	a.	Define CNF. Convert the following CFG to CNF $E \rightarrow E + T / T \quad T \rightarrow T * F / F \quad F \rightarrow (E) / I \quad I \rightarrow I a / I b \mid a \mid b$	10	L3	CO4
	b.	State and prove pumping lemma for context Free Grammars. Show that $L = \{ 0^n 1^n 2^n \mid n \geq 1 \}$ is not content free.	10	L3	CO4

OR

Q.8	a.	Define CNF convert the following CFG to CNF $S \rightarrow AB \quad A \rightarrow aA \mid bB \mid b \quad B \rightarrow b$	10	L3	CO4
	b.	Prove that the context free languages are closed under i) Union ii) Concatenation iii) Homomorphism	10	L3	CO4

Module – 5

Q.9	a.	Define a Turing Machine. Explain the working of a basic Turing machine with neat diagram.	08	L2	CO5
	b.	Design a Turing Machine to accept the language $L = \{ a^n b^n c^n \mid n \geq 1 \}$. Draw the transition diagram and show the moves made by TM for the string : aabbcc.	12	L3	CO5

OR

Q.10	a.	What are the programming Techniques for Turing Machine. Explain.	10	L2	CO5
	b.	Write short notes on : i) Multi Tape Turing Machine ii) Non Deterministic Turing Machine	10	L2	CO5